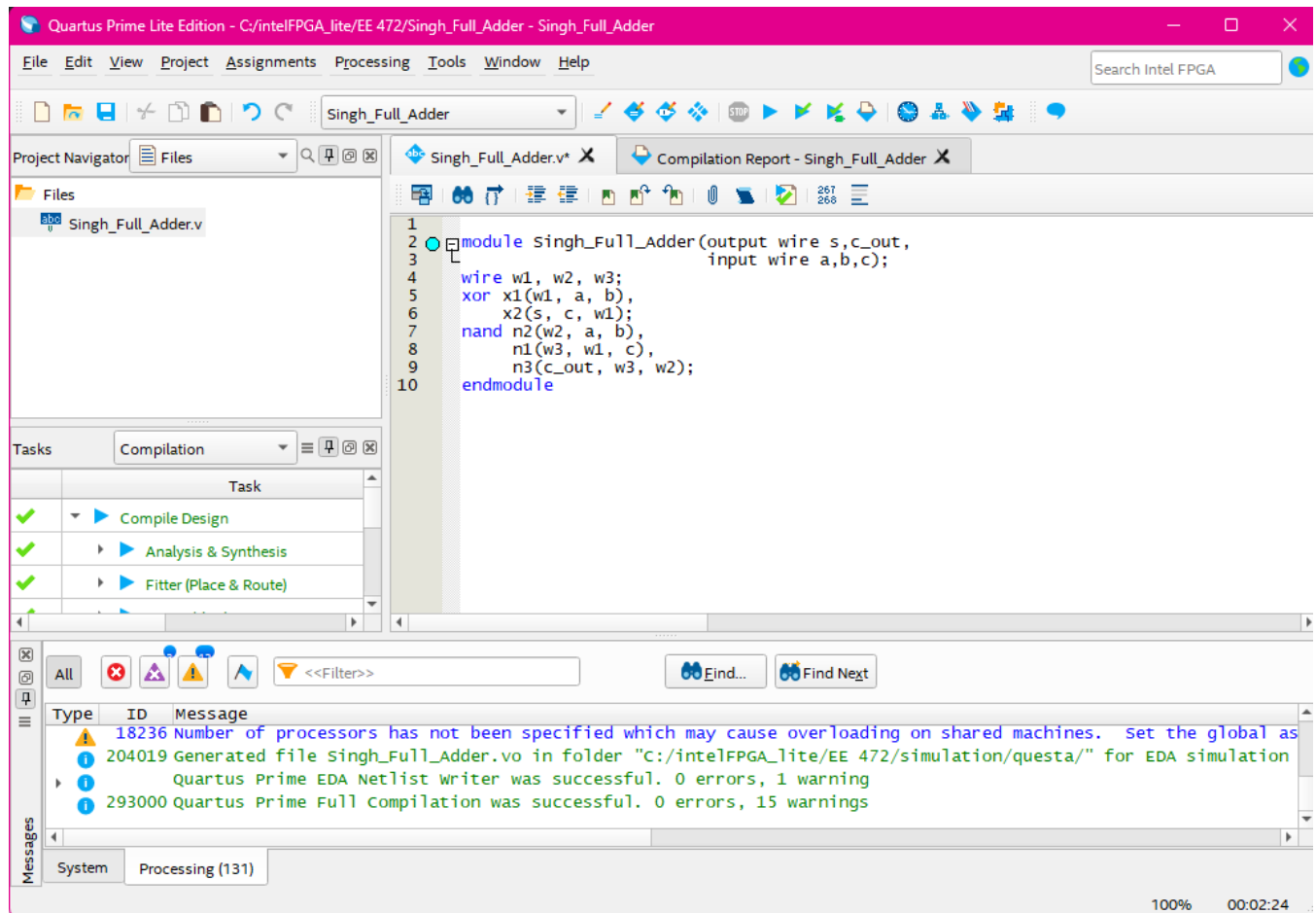


Annah Singh

EE 472

Homework #1

1. Full Adder Implementation & Compilation



2. XOR Compound Gate CMOS

$$A \oplus B = Y$$

$$0 \quad 0 \quad 0$$

$$0 \quad 1 \quad 1$$

$$1 \quad 0 \quad 1$$

$$1 \quad 1 \quad 0$$

$$Y = \overline{(AB)}(A+B)$$

$$0 = (\overline{0 \cdot 0})(0+0) \rightarrow 0 = (1)(0)$$

$$1 = (\overline{0 \cdot 1})(0+1) \rightarrow 1 = (1)(1)$$

$$1 = (\overline{1 \cdot 0})(1+0) \rightarrow 1 = (1)(1)$$

$$0 = (\overline{1 \cdot 1})(1+1) \rightarrow 0 = (0)(1)$$

$$Y = (A' + B')(A + B)$$

$$Y = A'A + A'B + AB' + B'B = A'B + AB'$$

$$Y = \overline{A'B + AB'}$$

$$Y = \overline{(A+B')(A'+B)}$$

nmos $\rightarrow$ and-series or-parallel

